

Synthetic Cultural Agents from Aggregate Anchors

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Abstract

Language models can score proposed survey items cheaply, but their answers mix supplied information with learned associations. We construct synthetic cultural agents from declared aggregate preference anchors. A shared bank of opposing responses receives deterministic labels from the signs of country scores, and Direct Preference Optimization fits a joint choice policy in adapter weights. Evaluation prompts omit country identifiers. In an application linking the Global Preferences Survey to World Values Survey items, the adapters retain trust anchor ordering more strongly than a country-named prompting comparator. Agreement with human country orderings remains uncertain and differs across dimensions. The construction makes the population-specific labeling rule inspectable, while leaving pretrained knowledge and synthetic-bank semantics in the policy. It provides a reusable scoring instrument whose prospective role in survey pre-piloting requires independent human validation.

Keywords: synthetic cultural agents; aggregate anchors; preference optimization; survey methodology; construct validity

1 Introduction

Group-level preference facts are expensive to obtain and easy to over-interpret. A proposed survey item must be piloted, translated, and fielded before researchers can assess whether it measures the intended attribute across populations. Language models offer a cheaper preliminary readout: researchers supply an item and a population description and obtain synthetic responses [1–3]. Yet sensitivity to prompting, compressed variance, and unstable subgroup comparisons limit the interpretation of those responses [4–6]. A useful methodological question is which population information the researcher explicitly supplies and what subsequent comparisons can establish.

We address this question by constructing *anchored synthetic choice policies*. A researcher declares aggregate preference benchmarks, uses their signs to label a common bank of synthetic contrasts, and fits a language-model adapter with Direct Preference Optimization (DPO) [7]. We interpret the loss through a regularized Bradley–Terry pairwise choice model. Each adapter is jointly trained across the preference dimensions. At evaluation, it assigns probabilities to supplied response codes for new items without a country identifier in the prompt. This restriction removes direct country-name conditioning at inference; pretrained knowledge, instruction tuning, generator judgments, and optimization variation remain.

The empirical application uses country scores from the Global Preferences Survey (GPS) [8]. These are aggregates of survey measures selected through prior experimental validation. They are

not country-level collections of incentivized experimental choices. We evaluate the fitted policies on candidate indicators in World Values Survey (WVS) Wave 7 [9]. Three country-level associations distinguish retention of the construction signal from agreement with people: human–anchor, policy–anchor, and policy–human. Held-out synthetic comparisons separately measure whether fitting recovers the imposed labels.

A *construct* is a theoretical preference attribute, a *measure* operationalizes that attribute through answers or choices, and an *anchor* is the resulting aggregate score used in construction. A fitted synthetic policy is a computational scoring instrument. None of these objects is interchangeable. In particular, aggregate scores do not identify a joint distribution of human preferences, and country-level associations do not establish individual-level measurement validity.

Our contributions are threefold:

1. We specify an inspectable route from declared aggregate preference signs to jointly fitted synthetic choice policies.
2. We evaluate this construction on a separate survey battery, distinguishing synthetic-label recovery, external anchor ordering, and human agreement.
3. We use discrepancies among these readouts to diagnose cross-survey coherence and to identify questions requiring further validation.

The intended application is group-level empirical analysis and survey pre-piloting, as a complement to human research. This application demonstrates text-responsive scoring and trait-dependent anchor transfer. Whether such scores improve item selection before a human pilot remains an empirical question.

2 Related research

Language models are used both to simulate survey responses and to adapt outputs to cultural communities. These purposes overlap technically, but the information supplied to a model and the evaluation target differ. Prompted synthetic respondents can resemble human samples on selected tasks [1, 2]; subsequent studies document sensitivity, compressed variation, and misportrayal of identity groups [4–6]. Prior synthetic cultural-agent work uses cultural descriptions to study behavior in experiments [10]. We extend this research agenda by making aggregate preference signs the explicit population-specific labeling input.

Prompted simulation supplies a population description at inference and reads associations already present in the model. A structural demand or simulated-method-of-moments exercise would invert a named preference parameter from observed choice probabilities. We compile declared aggregate signs into a scoring policy. The GPS–WVS comparisons report criterion associations for that construction.

Fine-tuning and cultural adaptation already occupy much of this methodological space. Table 1 identifies close comparisons. CultureLLM uses WVS seed data and semantic augmentation for culture-specific fine-tuning [11]. Cao et al. fit country-level survey response distributions and examine cross-survey evaluation [12]. SubPOP learns from subpopulation survey distributions and tests transfer to new survey settings [13]. These studies preclude a broad claim to first use cultural fine-tuning, aggregate survey information, or cross-survey evaluation.

Culturally grounded DPO is also established as an approach. Jung and Kim generate preference triplets around Korean legal and social norms and fit cultural response policies [14]. Our narrower distinction is the route from economic-preference signs through a shared pair bank to scoring on a separate instrument without a country name. Whether every element of this combination is unprecedented is not established by the present literature comparison.

Table 1: Selected methodological comparisons. The table describes inputs and targets; it is not an exhaustive priority claim.

Approach	Information supplied	Fitting or evaluation target
CultureLLM [11]	WVS seeds and semantic augmentation	Culture-specific adaptation
Cao et al. [12]	Country survey distributions	Distributional fit and cross-survey evaluation
SubPOP [13]	Subpopulation survey responses	Distribution prediction and transfer
Jung and Kim [14]	Korean cultural policy and synthetic triplets	DPO for cultural response coherence
Krsteski et al. [15]	Limited human responses	Allocation to synthesis and rectification
Present construction	Aggregate preference signs and a shared contrast bank	Joint adapters; country-unconditioned external scoring

The data requirement differs from direct fitting of a new survey’s response distributions: construction uses existing aggregate anchors and synthetic contrasts before human responses to the new instrument are available. That distinction does not imply that scarce human observations should always be spent on training. Krsteski et al. examine the roles of prompting, fine-tuning, and statistical rectification when human data are limited [15]. Our proposed pre-pilot use must therefore be compared with alternatives that allocate human measurement effort differently.

The econometric interpretation follows work treating language models as conditional response distributions [16], with DPO providing a reference-policy parameterization of Bradley–Terry comparisons [7]. The measurement interpretation requires more caution. Construct validity depends on cumulative evidence about the relation between a theoretical attribute and its measures [17]; cross-national applications also require attention to comparability [18, 19]. Agreement between an anchored policy and a survey may have several explanations, as may disagreement. We therefore report anchor retention and human agreement separately and treat the construction as a complement to human measurement.

3 Methodology

We separate the declared population input from the information already present in the models. The researcher chooses aggregate anchors, the synthetic contrast protocol, the fitting procedure, and the evaluation prompt. Declaring these inputs makes the conditioning rule inspectable. It does not restrict all information that can influence the fitted policy.

3.1 From aggregate anchors to synthetic comparisons

Let $z_c = (z_{c1}, \dots, z_{cK})$ contain the aggregate scores for group c . In this application, the coordinates describe trust, risk taking, patience, altruism, positive reciprocity, and negative reciprocity. The production protocol uses the binary profile $s_{ck} = 1\{z_{ck} \geq 0\}$. A teacher and a response generator supply a shared bank $B = \{(x_i, k_i, y_i^+, y_i^-)\}$, where x_i is a scenario and the two responses are intended to express higher and lower levels of dimension k_i . These intended loadings are generator judgments, rather than experimentally verified trait manipulations.

For each group, the deterministic label is

$$(y_{w,i}, y_{\ell,i}) = \begin{cases} (y_i^+, y_i^-), & z_{c,k_i} \geq 0, \\ (y_i^-, y_i^+), & z_{c,k_i} < 0. \end{cases} \quad (1)$$

The production relabeling path reuses fixed scenario and response text. Country identifiers and quantitative profile prose remain metadata; neither is supplied as a country-specific training prompt in this path. Profile rendering documents signs and magnitudes, but magnitudes do not affect these labels. Figure 1 shows this distinction. Appendix A gives an actual bank example, prompt excerpts, and source-backed pseudocode.

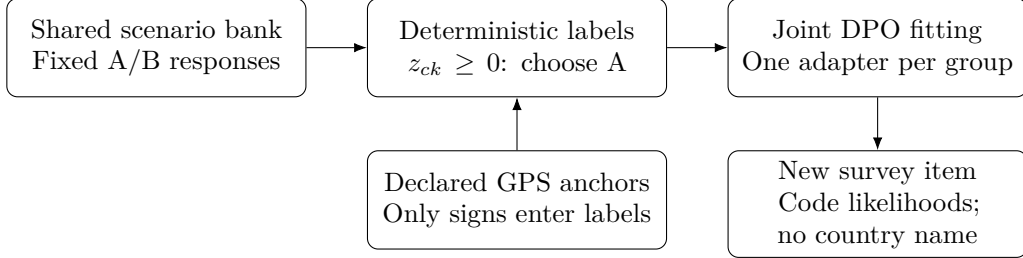


Figure 1: Production construction and readout. Profile prose and score magnitudes document the anchors but do not feed the deterministic label rule. The shared bank and the pretrained model also influence the resulting policy.

If two anchor vectors have identical signs, they induce identical labels on the same bank. Holding the split and fitting randomness fixed therefore leaves the fitting problem unchanged. Distinct fitted outputs among such groups cannot be interpreted as recovery of anchor magnitudes. Moreover, a dimension-specific evaluation does not isolate a dimension-specific adapter: all coordinates contribute to the same fitted policy. A changed sign vector defines an intervention on the construction algorithm only. It does not identify a human preference counterfactual.

Source-bank polarity. The United States is non-negative on all six GPS coordinates, so every training pair is labeled chosen A. Mexico is negative on all six, so every pair is labeled chosen B. On this bank, the USA–Mexico contrast is a global polarity flip of the same 658 scenarios.

The map from declared signs through the shared bank to a fitted policy is a construction. The researcher supplies z_c , applies the labeling rule, and obtains $\hat{\pi}_c$. Identification would require invertibility of a named parameter from observed WVS option probabilities. The present design does not recover z_c , the magnitudes $|z_{ck}|$, a single GPS coordinate, or a cultural utility. Rank association between adapter scores and GPS on held-out item text is criterion evidence for the construction. Whether fitted policies differ only as a function of the assigned sign vector remains a separate question.

3.2 DPO as a pairwise choice estimator

We treat a language model as a conditional distribution $\pi_\theta(y \mid x)$ over response sequences [16]. Under a Bradley–Terry model,

$$\Pr(y_w \succ y_\ell \mid x) = \sigma\{r(x, y_w) - r(x, y_\ell)\}, \quad \sigma(t) = (1 + e^{-t})^{-1}. \quad (2)$$

Independent type-I extreme-value utility shocks provide the usual random-utility interpretation [20]. Here the observed labels are deterministic synthetic choices; we use this likelihood as a fitting criterion rather than as an estimated model of human choice noise.

The KL-regularized policy problem maximizes expected reward minus $\beta D_{\text{KL}}(\pi \parallel \pi_{\text{ref}})$ for $\beta > 0$. On the support of the reference policy, its unrestricted optimum obeys

$$r(x, y) = \beta \log \frac{\pi^*(y \mid x)}{\pi_{\text{ref}}(y \mid x)} + \beta \log Z(x). \quad (3)$$

The input-specific additive term cancels from pairwise reward differences. Substitution gives the DPO loss [7]:

$$\Delta_\theta(x, y_w, y_\ell) = \log \frac{\pi_\theta(y_w | x)}{\pi_{\text{ref}}(y_w | x)} - \log \frac{\pi_\theta(y_\ell | x)}{\pi_{\text{ref}}(y_\ell | x)}, \quad (4)$$

$$\mathcal{L}_c(\theta) = -\frac{1}{|\mathcal{D}_c|} \sum_{(x, y_w, y_\ell) \in \mathcal{D}_c} \log \sigma\{\beta \Delta_\theta(x, y_w, y_\ell)\}. \quad (5)$$

This is a regularized Bradley–Terry interpretation through the reference-policy parameterization. It does not require an additional explicit KL term in the implemented loss. Finite synthetic comparisons constrain fitted margins on their support; they do not identify all responses, population marginals, or a unique human utility function.

QLoRA restricts the parameters that can move while keeping the quantized base fixed [21, 22]. We write the fitted policy as $\hat{\pi}_c = \pi_{\theta_0 + \Delta\theta_c}$. This efficient parameterization changes the feasible policy class, whereas the pairwise likelihood remains the DPO criterion. The unrestricted optimum in equation (3) need not be attained in this class. We retain Llama 3.1 8B Instruct because its limitations in prompted survey simulations provide a relevant setting for testing whether weight-embedded signals survive where prompting is weak [23].

3.3 Readout and empirical implications

For an item q , let a_{qj} be the token sequence of candidate response code j , including the prescribed leading space. We compute its teacher-forced log probability by summing conditional token log probabilities,

$$\ell_{cqj} = \sum_t \log \hat{\pi}_c(a_{qjt} | q, a_{qj, < t}), \quad \hat{P}_c(j | q) = \frac{\exp \ell_{cqj}}{\sum_h \exp \ell_{cqh}}. \quad (6)$$

The denominator is restricted to the supplied candidate codes. Code tokenization, leading whitespace, item wording, and option order are therefore part of the scoring specification. The forward pass requires an autoregressive model but no sampled answer; optional generated text is a separate output. The normalized code probabilities are a scoring convention whose dispersion need not match population choice frequencies.

For item recode g_q , the policy score is $\sum_j g_q(j) \hat{P}_c(j | q)$. Human scores use the same recode on valid answers before a survey-weighted average. We average item scores within each declared composite and compare country ranks. Appendix C records the item set, raw-code recodes, missingness rules, and aggregation choices. Group aggregation and aggregation across items are separate operations.

Held-out synthetic-label recovery asks whether the fitted relative margin improves over the reference, while raw choice accuracy asks whether the chosen response has higher likelihood. Policy–GPS rank association on new text asks whether anchor ordering transfers. Human–GPS association assesses the coherence of a proposed survey composite with the anchor. Policy–human association measures agreement with the survey’s country ordering. These are distinct empirical implications. Their joint pattern contributes evidence about measurement, but two correlations do not establish construct sharing or measurement invariance [17–19].

We assess distributional resemblance separately with

$$\text{TVD}_{cq} = \frac{1}{2} \sum_j |\hat{P}_c(j | q) - P_c^H(j | q)|. \quad (7)$$

A country-named persona comparator changes the information supplied at inference as well as the conditioning route. The weights-by-prompts comparison is consequently a configuration sensitivity

check. Isolating conditioning route would require, among other controls, a prompt receiving the same anchor information. Unconditioned evaluation removes direct country-name conditioning; it does not isolate anchor effects from every other source of variation.

4 Empirical application and results

We ask whether GPS-signed synthetic comparisons yield a policy that retains country ordering on a separate survey, and whether that ordering agrees with WVS respondents.

The diagnostic panel is Argentina, Brazil, China, Germany, Egypt, Great Britain, Greece, Indonesia, India, Japan, Mexico, Nigeria, the Netherlands, Russia, Turkey, and the United States. These sixteen countries are the complete census of adapters fitted in two production waves. The first wave covers China, Japan, Great Britain, the United States, Mexico, Argentina, Germany, and Russia. The second covers India, Indonesia, Nigeria, Egypt, Turkey, the Netherlands, Brazil, and Greece. We assembled those waves to include both global sign polarities, the three twin pairs that share a labeling problem, mixed profiles, and geographic spread. That is, the panel is a census of fitted production adapters. It is not a probability sample of the forty-two-country GPS–WVS intersection. A later study that treats country selection as a sampling problem should train every country in that intersection or a predeclared random subset of it.

The primary comparison holds countries, item sets, recodes, and aggregation rules fixed across arms. It contains twenty-three single-choice items, including nine trust items. Egypt has no valid Q69–Q71 observations and Great Britain has no valid Q174 observations, so these items are removed everywhere. Q12–Q14 are multiple-response items and are excluded from the categorical softmax evaluation. Table 2 lists the resulting map. Its proposed composites should not be read as validated measures of all six GPS traits. The original mapping preceded model scoring, whereas the strict common-support analysis and its sensitivity checks are retrospective.

Table 2: Candidate indicators on the strict common-support panel. Each row defines a candidate item set; latent-scale validity remains to be assessed. Patience items are interpreted separately.

Anchor	WVS items	Interpretation to be assessed
Trust	Q57–Q64, Q73	Generalized, interpersonal, and institutional trust/confidence
Patience	Q43; Q50	Less importance placed on work; financial satisfaction
Risk taking	Q106, Q107, Q109, Q178	Economic values and fare avoidance
Positive reciprocity	Q81	Confidence in charitable organizations
Negative reciprocity	Q176, Q177, Q179, Q195	Moral clarity and justifiability judgments
Altruism	Q99, Q101, Q103	Membership in environmental, charitable, and self-help groups

GPS supplies prior economic-preference benchmarks. Equivalence between each listed WVS item and its proposed GPS measure remains to be assessed. Q57 is the generalized-trust item related to the GPS validation comparison [8]; its human–GPS association in our sixteen-country panel is only 0.15. The wider trust composite pools substantively different items. The other mappings are exploratory bridges, especially where Wave 7 has no direct counterpart to the preference measure.

4.1 Recovery of the synthetic labels

Each country adapter uses 526 training comparisons from a shared 658-pair bank, with 132 held-out comparisons. The fitting configuration uses $\beta = 0.1$, LoRA rank 16 and scaling parameter 32, and one epoch. The countries instantiate thirteen distinct six-sign profiles (Table 3): Mexico shares its signs with Russia, India with Greece, and Indonesia with Egypt. Thus there are fewer distinct labeling problems than country labels. Appendix B records configuration and provenance limits.

Table 3: Thirteen distinct GPS sign profiles among the sixteen adapter countries. Coordinates are ordered trust, risk taking, patience, altruism, positive reciprocity, and negative reciprocity, matching Appendix A. The United States is non-negative on every coordinate; Mexico and Russia are negative on every coordinate. Identical sign vectors induce identical labels on the shared bank.

Members	trust	risk	patience	altruism	pos. recip.	neg. recip.
EGY, IDN	+	−	−	+	+	+
GRC, IND	−	−	−	−	−	+
MEX, RUS	−	−	−	−	−	−
ARG	−	+	−	+	+	−
BRA	−	−	−	+	+	−
CHN	+	−	+	+	+	+
DEU	−	−	+	−	−	−
GBR	+	+	+	+	−	+
JPN	−	−	+	−	−	+
NGA	−	+	−	−	−	+
NLD	+	+	+	−	−	+
TUR	+	+	−	−	−	+
USA	+	+	+	+	+	+

Pair-pooled across 2,112 held-out country–pair cells, relative-reward recovery is 0.942 and raw adapter choice accuracy is 0.819. The former records a positive improvement in the chosen–rejected margin over the reference; the latter records a positive adapted-model margin itself. Country-level relative-reward recovery ranges from 0.886 to 0.992. These results support recovery of imposed synthetic contrasts. They do not measure prediction of independently observed choices. Fifteen countries share the seed-42 holdout; Russia’s different 132-pair holdout shares 26 prompts with that set. This limits interpretation of pooled precision and country comparisons.

4.2 Cross-survey associations

Figure 2 compares human–GPS, adapter–GPS, adapter–human, and persona–human associations on the matched panel. Table 4 supplies the otherwise missing persona–GPS comparator. Country-bootstrap intervals use 2,000 common resamples and describe sensitivity conditional on the fitted policies, GPS anchors, WVS point estimates, and selected items. They do not include respondent sampling, anchor measurement error, or training randomness, and are not multiplicity-adjusted.

For trust, adapter–GPS association is 0.80 [0.55, 0.93], compared with −0.07 for the persona arm. The paired adapter-minus-persona difference is 0.87 [0.22, 1.36]. Human–GPS association on the sixteen-country rectangle is 0.39 [−0.18, 0.79], and adapter–human association is 0.33 [−0.23, 0.74]. The clearer finding is retention of anchor ordering on new trust text. The sixteen-country intervals leave human agreement unresolved, and they also leave the map’s own criterion association unresolved. On the same nine-item recodes, the forty-two-country GPS–WVS intersection yields human–GPS association 0.35 [0.09, 0.56] (Table 8). That larger human panel is a criterion check on the map.

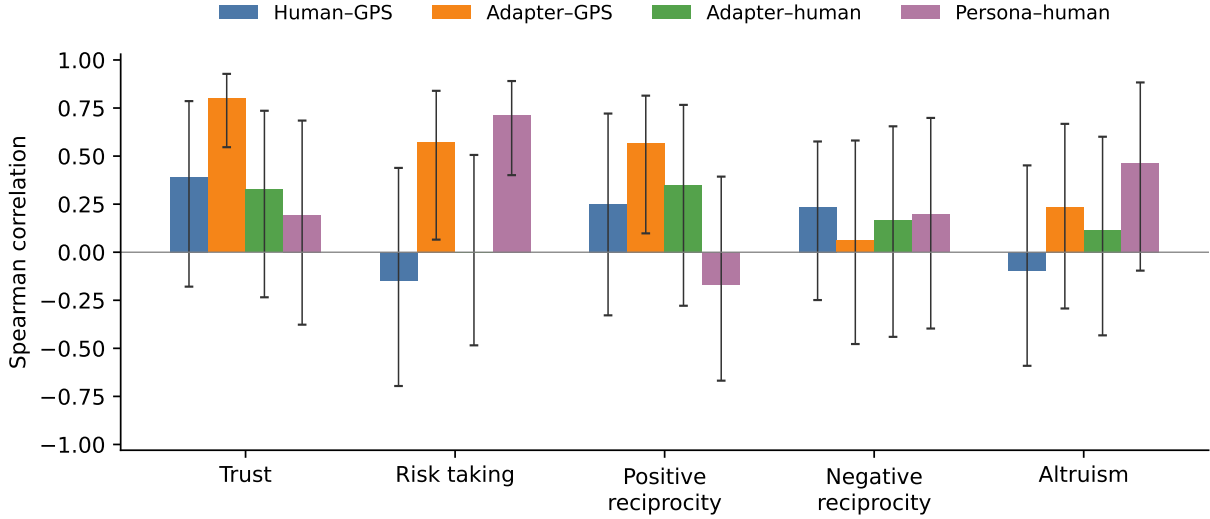


Figure 2: Strict sixteen-country panel: country-rank associations with 95% country-bootstrap intervals. Intervals exclude training-seed variation, respondent resampling, and GPS measurement error. The four series answer different questions; agreement with the anchor is distinct from agreement with human responses. Patience is omitted here because Q43 and Q50 have opposing human associations and is reported itemwise in Table 5.

Adapters are not retrained, and the adapter-human interval is unchanged. Similarity between a correlation and a product of two other correlations would not identify a mechanism, so we draw no such inference.

Table 4: Country-named persona-GPS association on the strict panel, with 95% country-bootstrap intervals. This supplies the anchor comparator for Figure 2.

Anchor	ρ	Interval
Trust	-0.07	[-0.55, 0.45]
Risk taking	-0.11	[-0.72, 0.51]
Positive reciprocity	0.02	[-0.47, 0.48]
Negative reciprocity	0.34	[-0.17, 0.82]
Altruism	-0.23	[-0.64, 0.26]

Risk taking shows why the targets must remain distinct. Adapter-GPS association is 0.57, human-GPS is -0.15 , and adapter-human is exactly zero in this sample, with interval $[-0.48, 0.51]$. The zero records zero sample rank association; it establishes neither independence nor different latent constructs. Persona-human association is 0.71 while persona-GPS is -0.11 . Mapping choices, anchor measurement, ecological differences, bank semantics, and fitting or scoring limitations can all contribute to this divergence. Positive reciprocity has only Q81 on the strict panel. Negative reciprocity and altruism likewise provide no resolved adapter-human result.

The two patience items have opposite human-GPS signs (Table 5). Q43 concerns reduced importance of work, whereas Q50 concerns financial satisfaction. We retain their original coding and report them separately rather than selecting a polarity after observing the correlations. Their

Table 5: Patience candidate indicators, separately scored on sixteen countries. Entries are Spearman correlations; brackets give 95% country-bootstrap intervals.

Pairing	Q43	Q50
Human-GPS	-0.56 [-0.86, -0.04]	0.62 [0.11, 0.91]
Adapter-GPS	0.01 [-0.47, 0.44]	0.44 [-0.12, 0.79]
Adapter-human	0.29 [-0.19, 0.73]	0.25 [-0.32, 0.70]
Persona-GPS	-0.62 [-0.90, -0.10]	0.62 [0.16, 0.87]
Persona-human	0.59 [0.11, 0.82]	0.70 [0.23, 0.91]

two-item mean is an available computational summary, but it is not treated as a validated patience measure. The main interpretation therefore does not use its aggregate correlation.

Retrospective influence checks leave trust adapter-GPS association between 0.779 and 0.850 when one country is omitted, and between 0.659 and 0.847 when one trust item is omitted. Resampling thirteen whole sign profiles gives 0.80 [0.45, 0.92]; equal weighting of profile means gives 0.85 [0.53, 0.98]. These are within-panel sensitivity checks. They do not demonstrate transport to new countries or independently generated scenario banks. Permuting the sixteen GPS trust scores across the frozen adapter composites, without refitting, yields a null median near zero and one-sided $p = 0.002$ (seed 20260820; 1,000 draws). That assignment check is not a sign-retrain test. Appendix C reports the paired Egypt and Q69-Q71 surfaces.

4.3 Distributional fit and prompt sensitivity

Table 6 compares configurations on the same twenty-three-item surface. Adding a persona prompt to an adapter yields trust-GPS association 0.67, below the unconditioned adapter’s 0.80. The persona baseline has lower TVD than either adapter configuration. Directional anchor retention and closeness to human response distributions therefore need not favor the same configuration.

Table 6: Configuration sensitivity on the strict panel. TVD averages 368 country-item cells. The adapter-persona row uses the space-prefixed rescore. The base TVD uses banked runs; no base country-ordering estimate is interpreted because run variation is confounded with bank. Rank associations on this table use the same country bootstrap as Figure 2, which excludes training-seed variation and GPS measurement error.

Configuration	Trust-GPS ρ	Mean TVD
Base, unconditioned	—	0.462
Base, persona	-0.07	0.369
Adapter, unconditioned	0.80	0.480
Adapter, persona	0.67	0.401

The pattern is dimension-dependent. With a country prompt, risk adapter-GPS association changes from 0.57 to -0.26 and adapter-human changes from zero to 0.62. These changes describe configuration sensitivity. They do not separate latent priors from anchor effects or establish a general division between weights determining direction and prompts determining shape. The historical base runs also vary numerically despite identical saved prompts. A single fixed base distribution would have undefined country-rank association on the common item set. Full checkpoint, tokenizer, and realized-prompt equivalence across runs remains a release gate.

The earlier rewrite also reported a broader thirty-item shape diagnostic. We do not mix its TVD levels with the strict-panel estimates, and omit its exhibit pending reconciliation of item eligibility

and historical aggregation. In particular, an ordering result alone cannot establish population dispersion. The evidence supports a fitted synthetic policy with trait-dependent external anchor association; reliable prediction of human country ranks and practical pre-pilot benefits remain unestablished.

5 Discussion

The construction makes one part of synthetic population conditioning explicit: the rule translating aggregate preference signs into training labels. A policy fitted to those labels can score new item text without naming the population in the evaluation prompt. This is a useful computational object for studying how supplied group information affects synthetic responses. The restriction applies to the declared conditioning channel; the model’s prior information and the semantic choices embedded in the shared bank remain part of the object.

The GPS–WVS application shows why construction success and external usefulness require separate evidence. Recovery of synthetic labels establishes that fitting moved the selected contrasts in the intended direction. Trust anchor ordering also transfers to a separate survey battery more strongly than in the country-named prompting comparator. Yet correspondence with human country ranks is uncertain. Proposed risk and patience indicators expose further discrepancies. We treat those discrepancies as questions about cross-survey coherence rather than as a diagnosis of which instrument failed. Convergent and discriminant evidence, together with cross-national comparability, would be needed to support a stronger construct-validity interpretation [17, 19]. GPS country scores are the declared anchor. Human–GPS association is the criterion for already-fielded item maps. On the forty-two-country intersection, the nine-item trust map tracks GPS in the expected direction, with an interval that excludes zero. Adapter–GPS association is retention of that anchor on new wording. Adapter–human association is agreement with people and remains unresolved on the sixteen-country panel. Consistency with published development or institutional gradients can screen a candidate measure. Those screens do not replace the human columns.

The method could be applied to regions, cohorts, or other groups with comparable aggregate anchors. That is scope of the construction, while demonstrated empirical transport is limited to the selected countries and item bank. Sign-only labeling discards anchor intensity and makes countries with common sign profiles share a labeling problem. Joint fitting also prevents attributing a dimension-specific output uniquely to that dimension’s coordinate. A magnitude-aware construction would be a different protocol; intensity weighting alone would not establish human predictive validity.

A prospective use is survey pre-piloting. Researchers could score candidate wording, compare its implied country ordering with declared anchors, and nominate items for independent human assessment. The present exercise does not establish that this process reduces cost, improves item selection, or shortens a questionnaire without losing validity. A blinded item-screening study should compare anchor-assisted selection with credible alternatives using a held-out human pilot. Appendix D records a prospective protocol for that study. Scoring one country or one item cannot certify measurement validity. Human observations also remain necessary for statistically justified rectification of synthetic estimates [15].

Several limitations bound the evidence. The country panel is a census of two production waves rather than a probability sample of the GPS–WVS intersection, repeated sign profiles reduce distinct construction inputs, and a single shared bank does not test sensitivity to bank generation. The shared bank is mixed-trait. Intended loadings are generator judgments rather than experimentally verified trait manipulations, so off-target movement on non-target facets remains a construction risk for any later dimension-specific reading. We do not apply a trait-isolation filter, and we do not treat

unused process diagnostics as evidence of purity. Prompt and checkpoint provenance differ across stored runs; bootstrap intervals condition on those runs rather than integrating their uncertainty. Neither pretraining exposure to survey material nor the accuracy of generator semantics is removed by country-token omission. The study also does not estimate effects of demographic or policy changes on human preferences. Perturbing an anchor tests the construction algorithm. Human causal claims require separate validation.

We intend the policies to complement field work. They should not be used to speak for individuals, replace confirmatory samples, or assign national cultural essences. A reusable scoring instrument can help formulate measurement questions, provided that its construction assumptions and unresolved validation tasks remain visible.

6 Conclusion

We construct synthetic cultural agents from declared aggregate preference anchors through a shared bank of sign-labeled comparisons. Joint adapter fitting and country-unconditioned scoring make the population-specific labeling rule inspectable. The resulting policy remains conditional on the pretrained model, bank semantics, and fitting procedure; it does not identify a human preference distribution.

In the GPS–WVS application, the adapters recover their synthetic labels and retain trust anchor ordering on new survey text more strongly than the country-named comparator. Human agreement remains uncertain, and the other dimensions provide mixed evidence. This distinction between anchor retention and survey prediction is central to interpreting the construction.

The prospective application is group-level survey pre-piloting and comparison of candidate indicators against declared anchors. Demonstrating that application requires an independent human item-screening study and stronger evidence of transport across banks and populations. Until then, the fitted policies are auditable scoring instruments intended to complement human survey research.

Code, data, and disclosures

The project repository is https://github.com/EconLLM-Lab/SCA2_PofW. Headline tables are regenerated from frozen country–item scores by `analysis/phase2/22_paper_a_surface.py`. Option-probability files used in scoring are posted as a zip of model outputs only, without survey microdata, at <https://drive.google.com/uc?export=download&id=1lIAx0ueSpgaZPmAznNbFSD31ddGQH7Nqo>. An unauthenticated fetch of that zip succeeded at the time of writing. Adapter weights reside in the Hugging Face repository `Bonorinoa/SCA2-phase2-adapters` and remain private until the authors make that repository public; they are optional for regenerating the locked tables and are required only to score new item text.

GPS and WVS data remain subject to their providers’ access and use conditions [8, 9]. This manuscript package does not redistribute respondent microdata. The analysis is at country level and should not be used to infer an individual’s preferences or to portray national populations as homogeneous. Independent human work would require its own consent and ethics procedures.

The study was not preregistered. The original draft describes an analysis plan written before outcome inspection; the common-support corrections, the forty-two-country human map, and reported sensitivity checks are retrospective. The supplied draft declares no funding and no competing interests; those declarations require author confirmation before submission. Language models are the scientific object of the study and were also used for writing and coding assistance. The authors remain responsible for the claims, analysis, and final manuscript.

References

- [1] Lisa P. Argyle, Ethan C. Busby, Nancy Fulda, Joshua R. Gubler, Christopher Rytting, and David Wingate. Out of one, many: Using language models to simulate human samples. *Political Analysis*, 31(3):337–351, 2023. doi: 10.1017/pan.2023.2.
- [2] John J. Horton, Apostolos Filippas, and Benjamin S. Manning. Large language models as simulated economic agents: What can we learn from homo silicus? Working Paper 31122, National Bureau of Economic Research, April 2023. Revised February 2026.
- [3] Trent D. Buskirk, Florian Keusch, Leah von der Heyde, and Adam Eck. More parameters than populations: A systematic literature review of large language models within survey research, 2026. URL <https://arxiv.org/abs/2509.03391>.
- [4] James Bisbee, Joshua D. Clinton, Cassy Dorff, Brenton Kenkel, and Jennifer M. Larson. Synthetic replacements for human survey data? the perils of large language models. *Political Analysis*, 32(4):401–416, 2024. doi: 10.1017/pan.2024.5.
- [5] Julien Boelaert, Samuel Coavoux, Étienne Ollion, Ivaylo Petev, and Patrick Präg. Machine bias. how do generative language models answer opinion polls? *Sociological Methods & Research*, 54(3):1156–1196, 2025. doi: 10.1177/00491241251330582.
- [6] Angelina Wang, Jamie Morgenstern, and John P. Dickerson. Large language models that replace human participants can harmfully misportray and flatten identity groups. *Nature Machine Intelligence*, 7:400–411, 2025. doi: 10.1038/s42256-025-00986-z.
- [7] Rafael Rafailov, Archit Sharma, Eric Mitchell, Stefano Ermon, Christopher D. Manning, and Chelsea Finn. Direct preference optimization: Your language model is secretly a reward model. In *Advances in Neural Information Processing Systems*, volume 36, 2023.
- [8] Armin Falk, Anke Becker, Thomas Dohmen, Benjamin Enke, David Huffman, and Uwe Sunde. Global evidence on economic preferences. *The Quarterly Journal of Economics*, 133(4): 1645–1692, 2018. doi: 10.1093/qje/qjy013.
- [9] World Values Survey Association. World values survey wave 7 (2017–2022). Country-pooled datafile. Madrid and Vienna: JD Systems Institute and WWSA Secretariat, 2022. URL <https://www.worldvaluessurvey.org/WVSDocumentationWV7.jsp>. Obtain from WWSA; we do not redistribute microdata. Former DOI 10.14281/18241.18 no longer resolves.
- [10] C. Mónica Capra, Augusto Gonzalez-Bonorino, and Emilio Pantoja. LLMs can model non-WEIRD populations: Experiments with synthetic cultural agents, 2025.
- [11] Cheng Li, Mengzhuo Chen, Jindong Wang, Sunayana Sitaram, and Xing Xie. CultureLLM: Incorporating cultural differences into large language models. In *Advances in Neural Information Processing Systems*, volume 37, 2024. URL https://proceedings.neurips.cc/paper_files/paper/2024/hash/9a16935bf54c4af233e25d998b7f4a2c-Abstract-Conference.html.
- [12] Yong Cao, Haijiang Liu, Arnav Arora, Isabelle Augenstein, Paul Röttger, and Daniel Herscovich. Specializing large language models to simulate survey response distributions for global populations. In *Proceedings of the 2025 Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*, pages 3141–3154, Albuquerque, New Mexico, 2025. Association for Computational Linguistics. doi: 10.18653/v1/2025.naacl-long.162.

- [13] Joseph Suh, Erfan Jahanparast, Suhong Moon, Minwoo Kang, and Serina Chang. Language model fine-tuning on scaled survey data for predicting distributions of public opinions. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 21147–21170. Association for Computational Linguistics, 2025. doi: 10.18653/v1/2025.acl-long.1028.
- [14] MinJae Jung and Minwoo Kim. Korean culture into LLM alignment: Toward cultural coherence, 2026. URL <https://arxiv.org/abs/2606.06797>.
- [15] Stefan Krsteski, Giuseppe Russo, Serina Chang, Robert West, and Kristina Gligorić. Valid survey simulations with limited human data: The roles of prompting, fine-tuning, and rectification. In *Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics*, pages 10887–10906. Association for Computational Linguistics, 2026. doi: 10.18653/v1/2026.acl-long.498.
- [16] Jens Ludwig, Sendhil Mullainathan, and Ashesh Rambachan. Large language models: An applied econometric framework. *Annual Review of Economics*, 18:283–316, 2026. doi: 10.1146/annurev-economics-120925-105620.
- [17] Donald T. Campbell and Donald W. Fiske. Convergent and discriminant validation by the multitrait-multimethod matrix. *Psychological Bulletin*, 56(2):81–105, 1959. doi: 10.1037/h0046016.
- [18] Jan-Benedict E. M. Steenkamp and Hans Baumgartner. Assessing measurement invariance in cross-national consumer research. *Journal of Consumer Research*, 25(1):78–107, 1998. doi: 10.1086/209528.
- [19] Eldad Davidov, Bart Meuleman, Jan Cieciuch, Peter Schmidt, and Jaak Billiet. Measurement equivalence in cross-national research. *Annual Review of Sociology*, 40:55–75, 2014. doi: 10.1146/annurev-soc-071913-043137.
- [20] Ralph Allan Bradley and Milton E. Terry. Rank analysis of incomplete block designs: I. the method of paired comparisons. *Biometrika*, 39(3/4):324–345, 1952. ISSN 00063444, 14643510. URL <http://www.jstor.org/stable/2334029>.
- [21] Edward J. Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. LoRA: Low-rank adaptation of large language models. In *International Conference on Learning Representations*, 2022.
- [22] Tim Dettmers, Artidoro Pagnoni, Ari Holtzman, and Luke Zettlemoyer. QLoRA: Efficient finetuning of quantized LLMs. In *Advances in Neural Information Processing Systems*, volume 36, 2023.
- [23] Ami Dalloul and Moritz Pfeifer. Can LLMs mimic household surveys? from representative agents to population distributions. SSRN 6703798, 2026.

A Construction protocol and worked example

The production manifest identifies the sign relabeling run as `gps_sign_relabel_all_20260816_004545Z`. Its source is the June 2026 USA–Mexico fixed-response bank. Relabeling deduplicates by dimension and prompt, rejects conflicting A/B text, and applies the sign rule to the recovered common bank. The exported countries exceed the trained panel; an exported label file does not imply that an adapter was fitted. The generation manifest sets `use_anchors=false`, so optional WVS anchors are not an input to this recorded source run.

The following pseudocode describes the production logic in `synthetic_generation/sca2_data_gen/relabel.py`, especially `build_triplet_bank`, `sign_choice`, and `label_triplet_for_country`. It is explanatory pseudocode rather than a claim that the complete replication package has been released.

```
Recover shared (scenario, dimension, response A, response B) bank.
Deduplicate on dimension and scenario; fail on conflicting A/B text.
For each group and each shared pair:
    Read finite GPS score for the pair's target dimension.
    Choose A when score >= 0; otherwise choose B.
    Export scenario, chosen text, rejected text; retain metadata.
Fit one adapter jointly on the group's training comparisons.
Score candidate codes for new items with the fixed evaluation prompt.
```

A.1 A real fixed-response contrast

The example below is triplet 00023ac74dac in `triplets_bank.jsonl`. The target is positive reciprocity. Argentina's exported score is +0.1596793085; its chosen option is A. This is an example of a deterministic training label. It does not describe an observed Argentine choice.

Scenario. A fellow student tutored me for free before a critical exam, and now they're facing a tight deadline for their own project and need someone to review their work. Decision: I can either take the time to thoroughly review their project or concentrate on my own study schedule. Trade-off: Helping my fellow student would acknowledge the effort they put into tutoring me, but it would cut into my own study time and potentially impact my performance.

Response A. I'll set aside time to carefully review their work this evening. They went out of their way to help me when it mattered most, and I want to make sure they know their effort made a difference. If I don't take this seriously, it would feel like I'm ignoring the support that helped me succeed.

Response B. I'll let them know I can only do a quick scan tomorrow morning. My exam prep schedule is already packed with material I haven't mastered yet, and I need to be realistic about what I can manage without burning out. I hope they understand my limitations in this situation.

The Argentina metadata records signs (−, +, −, +, +, −) in the order trust, risk taking, patience, altruism, positive reciprocity, negative reciprocity. Its policy is fitted jointly across these signs. A new WVS item, such as Q81 on confidence in charitable organizations, is then scored using the fitted adapter and the fixed code-likelihood readout. The scenario and Q81 are different texts. Their shared proposed dimension is an assumption of the map, and is not established by this example.

A.2 Profile documentation and generation prompts

The rendering function `build_cultural_profile` in `profiles.py` starts with "GPS CULTURAL STATE VECTOR (Falk et al. 2018, standardized deviations from global mean):". Each line has the

format “symbol (dimension) = signed score: magnitude. description”. It describes $|z| < 0.10$ as “near the global average”, $0.10 \leq |z| < 0.40$ as “moderately above/below average”, and larger values as “strongly above/below average”. Scores are printed to two decimals. These thresholds describe profile prose; they do not modify the production sign labels. A legacy model-selection path used profile prose, but it is distinct from the deterministic production path examined here.

The current generation source records the following exact scenario-template instructions:

Context: One realistic sentence establishing the agent, setting, and stakes.
 Decision: One sentence stating the two behaviorally plausible options the agent is choosing between.
 Trade-off: One sentence making the core target-facet tension explicit without naming high/low GPS scores.

The fixed-pair prompt in `generate.py` includes these exact requirements:

Generate two opposing responses to this same scenario.
 - Response A should load positively on the target dimension: it should express a higher level of the target trait/facet.
 - Response B should load negatively on the target dimension: it should express a lower level or absence of the target trait/facet.
 - Positive loading does not mean morally better, more polite, or more socially desirable.
 - Negative loading does not mean irrational, careless, hostile, or cartoonishly selfish.

Other instructions request matching perspective, tone, length, stakes, and the remaining traits, and prohibit country labels. They express design intentions rather than verified controls. Complete source prompt templates are transcribed in the accompanying `PROMPT_SOURCE_EXCERPTS.md`. Current source text is distinguished there from archived realized evaluation prompts; a complete historical request log tying every generation call to an immutable provider revision has not been verified.

The source manifest names endpoint aliases `hf-teacher`, `hf-generator`, and `hf-scorer`, with temperatures 0.7, 0.8, and 0.1. Current configuration maps them to Llama-3.3-70B-Instruct, Qwen3-32B, and Phi-4 endpoint names. Immutable served revisions remain unverified. The sixteen adapters examined here were produced by deterministic sign relabeling of the shared bank; the scorer was not a production gate. The manifest’s requested scenario count does not by itself explain the size of the recovered bank; we use the materialized bank and its 658 unique pairs as the production record.

A.3 Evaluation prompt and code scoring

The saved evaluation preview and `build_survey_user_prompt` agree on the unconditioned wrapper:

Answer this questionnaire as an individual person. Respond naturally and sincerely, as someone would in real life. Do not mention being an AI or assistant.
 Question:
[item text]
 Select exactly one response code from the options below.
[code: label, one option per line]
 Return only the response code, with no words or explanation.
 Answer:

For the persona condition, the first sentence becomes “Answer this questionnaire as a typical adult living in *[country name]*.” The tokenizer applies the chat template with a generation prompt. Candidate completion strings are a leading space followed by a code. Their log probabilities are

summed without length normalization, then normalized across codes at unit temperature. This is the readout in equation (6). Saved prompt previews include the rendered chat wrapper. The complete realized wrapper for Q57 is reproduced in the accompanying prompt document; it should be preserved when replaying the stored specification.

B Training configuration and historical diagnostics

The inspected CO2 multi-country notebook records Llama-3.1-8B-Instruct, 4-bit NF4 with double quantization and fp16 base compute, attention-projection LoRA modules `q_proj`, `k_proj`, `v_proj`, and `o_proj`, rank 16, scaling 32, and dropout 0.05. It uses DPO beta 0.1, learning rate 10^{-4} with cosine scheduling, one epoch, per-device batch size 1, and 16 gradient-accumulation steps. The maximum length is 768 with `keep_end` truncation. Trainable parameters are converted to fp32 after trainer initialization. The source has the warmup-ratio assignment commented out, so the prior draft’s claim that every adapter used warmup 0.03 is not retained.

These are inspected notebook settings. Their correspondence with every stored checkpoint remains to be established. The release package still needs an immutable checkpoint-to-run configuration manifest, exact dependency versions, and reference-log-probability provenance. Stored evidence supports the shared-bank 526/132 training/holdout counts and the Russia split exception reported in the main text. The notebook records a Tesla T4 run; account tier and historical billing have not been established, so no free-compute or cost claim is made.

A historical language-model diagnostic scored generated responses on all six GPS dimensions. It did not exclude pairs from production sign relabeling. Pair loadings remain generator judgments, and the shared bank is mixed-trait. Unused process diagnostics are not evidence of trait isolation. A new bank or a newly imposed exclusion rule would be a separate construction experiment.

C Measurement specification and sensitivity status

The strict panel uses Table 2’s common item set for every country and arm. We remove raw WVS missing codes -1 through -5 and nonfinite observations before any transformation, retaining valid zero membership responses. Human item means use positive `W_WEIGHT` weights and require at least fifty valid positive-weight responses per cell. All retained strict-panel cells pass that requirement in the saved audit.

The recodes inherited from the declared analysis are $g_{57}(v) = 1\{v = 1\}$; $g_q(v) = 5 - v$ for Q58–Q64, Q73, and Q81; and $g_q(v) = 11 - v$ for Q177 and Q179. Other retained items use their raw codes. For a model, each code is recoded before taking its probability-weighted expectation. For humans, each valid answer is recoded before its weighted mean. Proposed dimension composites are unweighted averages of these item means, without standardizing their ranges. This makes scale ranges part of the composite definition. Unit-range recoding is a separate sensitivity specification in the saved audit; the reported surface retains its original scales.

The historic Q69–Q71 bridge recoded negative missing values before masking. That can turn missingness into valid-looking positive scores. Those items do not enter the strict panel. Earlier broad-panel institutional-trust coefficients from that faulty sequence are retired. A corrected broader human-only comparison uses a different country sample and must not be substituted for a matched model–human comparison. We retain Egypt in the primary sample. Table 7 separates Egypt removal from restoration of Q69–Q71 on the remaining fifteen countries, using frozen item scores and no adapter refit. Adapter–GPS trust remains between 0.79 and 0.81 across those surfaces. Human–GPS association moves more when the institutional items are restored.

Table 7: Paired missingness check for Egypt and Q69–Q71. Entries are Spearman correlations between country trust composites and GPS trust z . The primary panel retains Egypt and omits Q69–Q71 everywhere because those items were not asked in Egypt. Adapters were not refit.

Surface	Adapter–GPS	Human–GPS
16 countries, 9 trust items	0.80	0.39
15 countries (Egypt omitted), 9 items	0.81	0.47
15 countries, 12 items (Q69–Q71 restored)	0.79	0.59

The main intervals use common country-bootstrap draws (seed 20260905; 2,000 draws), average ranks for ties, and undefined values for constant vectors. Sign-profile resampling retains every member of each selected profile; the equal-profile analysis instead averages within profiles before ranking. Leave-country-out and leave-item-out results describe influence on this observed panel. None of these intervals includes repeated model fitting, respondent resampling, or GPS measurement error. An assignment permutation of the sixteen GPS trust z values across the frozen adapter composites (seed 20260820; 1,000 draws) yields a real Spearman of 0.80, a null median of 0.00, and one-sided $p = 0.002$. Adapters were not refit, and the test does not retrain on permuted sign labels. Retrained anchor-permutation experiments, new scenario banks, repeated training seeds, and independent survey batteries remain unrun validation tasks.

The same recodes on the forty-two-country GPS–WVS intersection supply a human-only criterion check on the map (Table 8). Adapter scores are not recomputed in that table.

Table 8: Human–GPS country-rank associations on the Paper A item map, using the same recodes as the sixteen-country rectangle. Rows require at least fifty valid weighted responses per item. Complete-item rows retain only countries observed on every listed item. Intervals are 95% country bootstrap (2,000 draws) on the human panel and omit GPS measurement error.

Surface	n	ρ	Interval
9 trust items (complete)	42	0.35	[0.09, 0.56]
Q69–Q71 only (complete)	41	0.45	[0.15, 0.67]
12 trust items including Q69–Q71	41	0.43	[0.14, 0.64]
Q43 vs GPS patience	42	−0.53	[−0.74, −0.27]
Q50 vs GPS patience	42	0.39	[0.09, 0.63]
Risk-taking map (complete)	41	−0.06	[−0.43, 0.31]
Positive reciprocity Q81	42	0.05	[−0.27, 0.39]
Negative reciprocity map	42	0.42	[0.17, 0.59]
Altruism map (complete)	42	0.03	[−0.31, 0.35]

D Prospective human item-screening protocol

This manuscript does not field a new survey. The intended use of the fitted policies is to score candidate item text against declared GPS anchors before a human pilot. The following protocol is recorded so that a later study can test that use without changing the present construction.

A researcher predeclares a battery of candidate items, their recodes, and which GPS coordinate each item is proposed to track. Frozen adapters from the present study score those items with the unconditioned space-prefixed code-likelihood readout. Country composites are then ranked against GPS z . Interpretation is comparative. Items that recover GPS order about as strongly

as the nine-item trust map does in the forty-two-country human panel are retained as aligned candidates. Items that recover GPS order strongly where the corresponding human association on already-fielded items is near zero are set aside as likely echo of the training signs. Items that fail to recover GPS order where the human map is strong are set aside as non-transporting.

A human check then fields a small number of retained and flagged items in a predeclared set of countries, using the same recodes. The confirmatory estimand is human–GPS country-rank association on the new items, with a secondary adapter–human association. The study does not treat adapter–GPS alone as construct validation. Ethics review, consent, and sample-size planning belong to that later study.